

Qualification Pack



Mining Supervisor (Mineral Sands)

QP Code: MIN/Q1805

Version: 1.0

NSQF Level: 5.5

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MIN/Q1805: Mining Supervisor (Mineral Sands)

Brief Job Description

A Mining Supervisor (Mineral Sands) supervises mine workers and coordinates various mine activities, including organizing and monitoring workflow for various processes. The individual is responsible for ensuring compliance with the applicable regulatory requirements and safety procedures at mining sites.

Personal Attributes

This job requires the individual to concentrate on the task at hand and complete it efficiently and effectively. The individual should also be physically agile, strong, have good eye sight and not suffer from colour-blindness.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [MIN/N1814: Supervise excavation, accumulation, loading and transportation of raw sand](#)
2. [MIN/N1815: Supervise the safety of dredging operations at DWUP](#)
3. [MIN/N1816: Supervise the conveyance of mining output through slurry pipelines](#)
4. [MIN/N1810: Follow the applicable health, safety and environmental guidelines \(Placer Mining\)](#)
5. [MIN/N1811: Work collaboratively and effectively in a team](#)
6. [DGT/VSQ/N0103: Employability Skills \(90 Hours\)](#)

Qualification Pack (QP) Parameters

Sector	Mining
Sub-Sector	Mining Operation
Occupation	Placer Mining
Country	India
NSQF Level	5.5
Credits	18

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Aligned to NCO/ISCO/ISIC Code	NCO-2015/ 3121.9900
Minimum Educational Qualification & Experience	Completed 3 year UG degree (i.e. Completed 3rd year of 3-year/ 4-year UG (Mining/ Mechanical/ Electrical)) OR Completed 2nd year diploma after 12th (Mining/ Mechanical/ Electrical) (with 1 year relevant experience) OR Completed 3 year diploma after 10th (Mining/ Mechanical/ Electrical) (with 2 years relevant experience) OR 12th grade Pass (with 3-years relevant experience) OR Previous relevant Qualification of NSQF Level (5.0)(with 1.5-year relevant experience) OR Previous relevant Qualification of NSQF Level (4.5)(with 3 years relevant experience)
Minimum Level of Education for Training in School	Not Applicable
Pre-Requisite License or Training	NA
Minimum Job Entry Age	18 Years
Last Reviewed On	NA
Next Review Date	14/03/2027
NSQC Approval Date	15/03/2024
Version	1.0
Reference code on NQR	QG-5.5-MG-02177-2024-V1-SCMS
NQR Version	1.0

Remarks:

540 Hours

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MIN/N1814: Supervise excavation, accumulation, loading and transportation of raw sand

Description

This occupational standards unit is about supervising/ coordinating different activities involving mineral production at the point of extraction, in accordance with the applicable requirements. It also includes supervising the use of Heavy Earth Moving Machinery (HEMM)

Scope

The scope covers the following :

- Protect personnel working in vicinity of site excavation
- Oversee the HEMM operations
- Oversee inspection and maintenance of branch road & main haul road to excavation site

Elements and Performance Criteria

Protect personnel working in vicinity of site excavation

To be competent, the user/individual on the job must be able to:

- PC1.** identify and prevent unauthorized entry at the site of excavation, and report it to the relevant official
- PC2.** check for the deployment of inexperienced and unsupervised personnel at the excavation site and report it to the relevant official
- PC3.** identify deficiencies in safety equipment and conditions in the mine or section and report to the relevant official
- PC4.** supervise hazardous operations to ensure they are carried out safely with the protection of personnel involved
- PC5.** ensure effective monitoring of the area under charge until the end of the shift or until relieved by a duly appointed substitute and acquainting with the matters requiring attention
- PC6.** supervise the personnel within the area of responsibility
- PC7.** ensure the personnel are able to identify hazardous occurrences situation and are aware of the appropriate action to be taken while taking ensuring own and other's safety
- PC8.** conduct appropriate checks on the competency of persons employed in mineral extraction
- PC9.** supervise the workers during work to ensure they do not come in contact with moving plant/ equipment/ machine parts
- PC10.** ensure the personnel use the appropriate Personal Protective Equipment (PPE) while conducting hazardous activities
- PC11.** check and ensure the personnel are not exposed to excessive work hazards (e.g. vibration, hazardous substances)
- PC12.** instruct the workers to use mechanical lifting devices to minimize manual handling tasks
- PC13.** inspect and ensure appropriate checks on safety devices are carried out
- PC14.** identify different problems, such as obstructions, geological movements, mechanical or service failure, and report to appropriate person for appropriate remedial action

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Oversee the HEMM operations

To be competent, the user/individual on the job must be able to:

- PC15.** monitor the operational cycle and all relevant operations within the area of responsibility
- PC16.** check if the material needs for the production unit are met using the material transport system and report any inconsistencies
- PC17.** conduct time study of the HEMM equipment to prevent idling of tipper/equipment
- PC18.** check for the prompt refilling of the excavated/ mined out areas
- PC19.** co-ordinate work operations within the area of responsibility to ensure each activity is carried out appropriately
- PC20.** ensure HEMM operations do not affect the health and safety of personnel
- PC21.** maintain appropriate documentation concerning the HEMM operations, following the organizational procedures

Oversee inspection and maintenance of branch road & main haul road to excavation site

To be competent, the user/individual on the job must be able to:

- PC22.** ensure the appropriate size and specification of the haul road
- PC23.** ensure appropriate illumination at the work site and haul road as per the applicable standards
- PC24.** supervise the implementation of applicable traffic rules on the haul road to prevent over-speeding and any accidents
- PC25.** inspect the loading of tippers to prevent overloading
- PC26.** check and ensure transportation of dry and clean material from the excavation site to the plant

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** the duties and responsibilities of a mining mate/supervisor as mentioned in Metalliferous Mines Regulations, 1961 and updated Directorate General Of Mines Safety (DGMS) circulars
- KU2.** the safety policy of IREL(India) Limited
- KU3.** Mining Rules and Regulations
- KU4.** the role of workmen under the Mines Act
- KU5.** the basic technology used in and the functioning of various components and equipment in dredge mining, and supplementary mining plant
- KU6.** the use of appropriate tools to assess and confirm technical faults in excavation equipment
- KU7.** the use of pressure indicators, fuel pressure testers, manifold gauge sets, oil pressure gauges, measuring equipment (Venire calipers, micrometer, feeler gauges, multi-meter, flow meter, temp gauge, dial gauge etc.), condition monitoring systems for temperature, vibration, sound etc.
- KU8.** the appropriate safety measures to be taken in the vicinity of different types of machinery
- KU9.** the importance of being aware of and following the locally prepared emergency response plan/ disaster management plan
- KU10.** the importance of promptly reporting the identified problems/ accidents/ incidents

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- KU11.** the impact and cost to the organization due to equipment breakdown or damage
- KU12.** the direct/ indirect costs and implications of accidents, delays in process, and violation of safety procedures for the organization
- KU13.** the conditions and work environment in a mine
- KU14.** the importance of following the equipment daily maintenance checklist
- KU15.** the appropriate precautions to be taken when handling compressor, pumps, dryers, spirals, and trommel
- KU16.** the maintenance procedures for dust suppression systems
- KU17.** the occupational diseases/hazards likely in the mining environment
- KU18.** the functioning and use of hydraulic systems - different types of pumps and control valves
- KU19.** the relevant mining standards and procedures followed in the company in detail
- KU20.** the importance of monitoring the mine operations in continuous succession of shifts by a competent person

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** carry out appropriate documentation concerning work
- GS2.** read the equipment manuals, process documents and internal information memos
- GS3.** follow the instructions of Original Equipment Manufacturer (OEM) manuals
- GS4.** communicate politely and professionally
- GS5.** listen attentively to understand the instructions or information being shared
- GS6.** plan and carry out different activities appropriately, following the task lists and schedules
- GS7.** ensure timely completion of work in compliance with the applicable quality standards
- GS8.** make quick decisions to deal with workplace emergencies
- GS9.** identify potential problems related to work and take appropriate preventive measures
- GS10.** evaluate the possible solutions to work-related problems applying reasoning skills and select the best one
- GS11.** organize the operations and service manuals to allow easy access to them
- GS12.** follow the organizational procedure during emergencies
- GS13.** identify the areas of improvement and take appropriate measures

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Protect personnel working in vicinity of site excavation</i>	16	26	-	10
PC1. identify and prevent unauthorized entry at the site of excavation, and report it to the relevant official	1	2	-	1
PC2. check for the deployment of inexperienced and unsupervised personnel at the excavation site and report it to the relevant official	3	2	-	1
PC3. identify deficiencies in safety equipment and conditions in the mine or section and report to the relevant official	1	2	-	1
PC4. supervise hazardous operations to ensure they are carried out safely with the protection of personnel involved	1	2	-	1
PC5. ensure effective monitoring of the area under charge until the end of the shift or until relieved by a duly appointed substitute and acquainting with the matters requiring attention	1	2	-	1
PC6. supervise the personnel within the area of responsibility	1	1	-	-
PC7. ensure the personnel are able to identify hazardous occurrences situation and are aware of the appropriate action to be taken while taking ensuring own and other's safety	1	1	-	1
PC8. conduct appropriate checks on the competency of persons employed in mineral extraction	1	2	-	1
PC9. supervise the workers during work to ensure they do not come in contact with moving plant/ equipment/ machine parts	1	2	-	-
PC10. ensure the personnel use the appropriate Personal Protective Equipment (PPE) while conducting hazardous activities	1	2	-	1

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC11. check and ensure the personnel are not exposed to excessive work hazards (e.g. vibration, hazardous substances)	1	2	-	1
PC12. instruct the workers to use mechanical lifting devices to minimize manual handling tasks	1	2	-	-
PC13. inspect and ensure appropriate checks on safety devices are carried out	1	2	-	-
PC14. identify different problems, such as obstructions, geological movements, mechanical or service failure, and report to appropriate person for appropriate remedial action	1	2	-	1
<i>Oversee the HEMM operations</i>	8	14	-	6
PC15. monitor the operational cycle and all relevant operations within the area of responsibility	1	2	-	1
PC16. check if the material needs for the production unit are met using the material transport system and report any inconsistencies	2	3	-	1
PC17. conduct time study of the HEMM equipment to prevent idling of tipper/equipment	1	1	-	1
PC18. check for the prompt refilling of the excavated/ mined out areas	1	2	-	1
PC19. co-ordinate work operations within the area of responsibility to ensure each activity is carried out appropriately	1	2	-	-
PC20. ensure HEMM operations do not affect the health and safety of personnel	1	2	-	1
PC21. maintain appropriate documentation concerning the HEMM operations, following the organizational procedures	1	2	-	1
<i>Oversee inspection and maintenance of branch road & main haul road to excavation site</i>	6	10	-	4
PC22. ensure the appropriate size and specification of the haul road	1	2	-	1



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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC23. ensure appropriate illumination at the work site and haul road as per the applicable standards	1	2	-	1
PC24. supervise the implementation of applicable traffic rules on the haul road to prevent over-speeding and any accidents	1	2	-	-
PC25. inspect the loading of tippers to prevent overloading	1	2	-	1
PC26. check and ensure transportation of dry and clean material from the excavation site to the plant	2	2	-	1
NOS Total	30	50	-	20

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National Occupational Standards (NOS) Parameters

NOS Code	MIN/N1814
NOS Name	Supervise excavation, accumulation, loading and transportation of raw sand
Sector	Mining
Sub-Sector	Mining Operation
Occupation	Placer Mining
NSQF Level	5.5
Credits	5
Version	1.0
Last Reviewed Date	15/03/2024
Next Review Date	14/03/2027
NSQC Clearance Date	15/03/2024

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MIN/N1815: Supervise the safety of dredging operations at DWUP

Description

This unit is about supervising the safety of dredging operations at the Dredge and Wet Upgradation Plant (DWUP).

Scope

The scope covers the following :

- Ensure safety in the dredging operations
- Implement the equipment safety measures

Elements and Performance Criteria

Ensure safety in the dredging operations

To be competent, the user/individual on the job must be able to:

- PC1.** ensure safe access to the float line pontoon
- PC2.** check the approach road is fitted with suitable gradient to float line pontoon
- PC3.** ensure the provision of handrails, chains, flaps on the float line pontoon, and arrange for their regular maintenance and painting
- PC4.** ensure appropriate illumination standards on the float line pontoon
- PC5.** direct the workers to shift material from the shore to the plant deck by using the maintenance barge
- PC6.** ensure no material is shifted using a single pontoon
- PC7.** direct the relevant personnel to conduct regular cleaning of the concentrator deck, head feed bin deck, and dredge deck
- PC8.** check for the regular cleaning of the manhole compartment, including its dewatering as per standard procedure
- PC9.** ensure regular cleaning of the spirals
- PC10.** monitor the free board draft of the concentrate deck, head feed bin deck, and dredge deck
- PC11.** carry out appropriate anchoring of the floating units, i.e. concentrator deck, head feed bin deck and dredge deck
- PC12.** check and ensure safe operating parameters of the dredge plant and concentrator plant are maintained
- PC13.** dispose of tailing with proper density with timely replacement of tails nozzle
- PC14.** conduct surface preparation of dredge face appropriately with the help of HEMM
- PC15.** carry out timely shifting of all the field setup
- PC16.** ensure the provision of safe and wholesome drinking water to the dredge plant

Implement the equipment safety measures

To be competent, the user/individual on the job must be able to:

- PC17.** ensure the use of appropriate safety gear for anchor block shifting, winch wire rope handling, locking of winches, sandblasting, etc.

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- PC18.** ensure the workers maintain a safe distance from the pond edge in front of the dredge and avoid walking on the pontoons during the movement of dredge
- PC19.** implement the Safety Management Plan (SMP) and Emergency Management Plan (EMP)
- PC20.** check and ensure the dredge deck floor is free of oil, grease etc.
- PC21.** ensure the workers follow the appropriate safety measures while working at haul roads, heights, dumps, sump area, stockyards, near moving parts, etc.
- PC22.** ensure no unauthorized personnel is present on the maintenance barge during material handling
- PC23.** ensure the workers follow the standard signal while lifting materials
- PC24.** ensure coupling guards/ belt guards of pumps are in the correct position
- PC25.** ensure the lighting distribution board is switched off while cleaning the spiral with a water jet
- PC26.** instruct the workers to use appropriate chain pulley, slings and ropes for loading and unloading materials
- PC27.** implement appropriate measures to prevent the workers from coming in contact with noxious fumes/ gases
- PC28.** supervise the dewatering inside the deck compartment to ensure it is carried out safely
- PC29.** ensure the use of headlamp of the recommended specification with proper gauge protection during deck cleaning
- PC30.** direct the workers maintain the pontoon handrails in good condition
- PC31.** ensure the use of caution and safety signs, stickers, and tags

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** different types of mines
- KU2.** the importance of discipline and punctuality in mine operations
- KU3.** the applicable standing orders in the mine
- KU4.** the applicable safety regulations
- KU5.** the provision of wages, working hours and accident compensation as per the Mines Act.
- KU6.** the processes, such as procurement, store management, inventory management, quality management and key contact points for query resolution
- KU7.** the applicable quality norms and standards for painting
- KU8.** the application of 5S principles in mining operations
- KU9.** the impact of rise and fall in air temperature on the physical/atmospheric working conditions
- KU10.** the effect of atmospheric temperature and dust/hazardous materials on mine workers
- KU11.** the use of appropriate Personal Protective Equipment (PPE) in mining operations
- KU12.** the maintenance of PPE

Generic Skills (GS)

User/individual on the job needs to know how to:



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- GS1.** carry out appropriate documentation concerning work
- GS2.** read the equipment manuals, process documents and internal information memos
- GS3.** follow the instructions of Original Equipment Manufacturer (OEM) manuals
- GS4.** communicate politely and professionally
- GS5.** listen attentively to understand the instructions or information being shared
- GS6.** plan and carry out different activities appropriately, following the task lists and schedules
- GS7.** ensure timely completion of work in compliance with the applicable quality standards
- GS8.** make quick decisions to deal with workplace emergencies
- GS9.** identify potential problems related to work and take appropriate preventive measures
- GS10.** evaluate the possible solutions to work-related problems applying reasoning skills and select the best one
- GS11.** organize the operations and service manuals to allow easy access to them
- GS12.** follow the organizational procedure during emergencies
- GS13.** identify the areas of improvement and take appropriate measures

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Ensure safety in the dredging operations</i>	15	27	-	10
PC1. ensure safe access to the float line pontoon	1	1	-	-
PC2. check the approach road is fitted with suitable gradient to float line pontoon	1	2	-	1
PC3. ensure the provision of handrails, chains, flaps on the float line pontoon, and arrange for their regular maintenance and painting	1	1	-	-
PC4. ensure appropriate illumination standards on the float line pontoon	1	1	-	-
PC5. direct the workers to shift material from the shore to the plant deck by using the maintenance barge	1	2	-	1
PC6. ensure no material is shifted using a single pontoon	-	2	-	-
PC7. direct the relevant personnel to conduct regular cleaning of the concentrator deck, head feed bin deck, and dredge deck	1	2	-	1
PC8. check for the regular cleaning of the manhole compartment, including its dewatering as per standard procedure	1	2	-	1
PC9. ensure regular cleaning of the spirals	1	1	-	-
PC10. monitor the free board draft of the concentrate deck, head feed bin deck, and dredge deck	1	2	-	1
PC11. carry out appropriate anchoring of the floating units, i.e. concentrator deck, head feed bin deck and dredge deck	1	2	-	1
PC12. check and ensure safe operating parameters of the dredge plant and concentrator plant are maintained	1	2	-	1
PC13. dispose of tailing with proper density with timely replacement of tails nozzle	1	2	-	1

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC14. conduct surface preparation of dredge face appropriately with the help of HEMM	1	2	-	1
PC15. carry out timely shifting of all the field setup	1	2	-	1
PC16. ensure the provision of safe and wholesome drinking water to the dredge plant	1	1	-	-
<i>Implement the equipment safety measures</i>	15	23	-	10
PC17. ensure the use of appropriate safety gear for anchor block shifting, winch wire rope handling, locking of winches, sandblasting, etc.	1	1	-	-
PC18. ensure the workers maintain a safe distance from the pond edge in front of the dredge and avoid walking on the pontoons during the movement of dredge	1	1	-	-
PC19. implement the Safety Management Plan (SMP) and Emergency Management Plan (EMP)	1	2	-	1
PC20. check and ensure the dredge deck floor is free of oil, grease etc.	1	2	-	1
PC21. ensure the workers follow the appropriate safety measures while working at haul roads, heights, dumps, sump area, stockyards, near moving parts, etc.	1	1	-	1
PC22. ensure no unauthorized personnel is present on the maintenance barge during material handling	1	1	-	1
PC23. ensure the workers follow the standard signal while lifting materials	1	1	-	1
PC24. ensure coupling guards/ belt guards of pumps are in the correct position	1	1	-	1
PC25. ensure the lighting distribution board is switched off while cleaning the spiral with a water jet	1	1	-	-
PC26. instruct the workers to use appropriate chain pulley, slings and ropes for loading and unloading materials	1	2	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC27. implement appropriate measures to prevent the workers from coming in contact with noxious fumes/ gases	1	2	-	1
PC28. supervise the dewatering inside the deck compartment to ensure it is carried out safely	1	2	-	1
PC29. ensure the use of headlamp of the recommended specification with proper gauge protection during deck cleaning	1	2	-	1
PC30. direct the workers maintain the pontoon handrails in good condition	1	2	-	-
PC31. ensure the use of caution and safety signs, stickers, and tags	1	2	-	1
NOS Total	30	50	-	20

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National Occupational Standards (NOS) Parameters

NOS Code	MIN/N1815
NOS Name	Supervise the safety of dredging operations at DWUP
Sector	Mining
Sub-Sector	Mining Operation
Occupation	Placer Mining
NSQF Level	5.5
Credits	4
Version	1.0
Last Reviewed Date	15/03/2024
Next Review Date	14/03/2027
NSQC Clearance Date	15/03/2024

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MIN/N1816: Supervise the conveyance of mining output through slurry pipelines

Description

This occupational standards unit is about supervising the conveyance of the dredged-out material from the dredging face to the processing plant for upgradation of heavy minerals through spiral separation

Scope

The scope covers the following :

- Supervise the conveyance of dredged output sand through pipelines

Elements and Performance Criteria

Supervise the conveyance of dredged output sand through pipelines

To be competent, the user/individual on the job must be able to:

- PC1.** check for the correct functioning of booster pumps added to combined slurry transportation system of mining output
- PC2.** ensure proper monitoring of ampere meter reading and discharge pressure of all booster pumps including pump attached to the output transfer bins
- PC3.** ensure proper monitoring of the water level of all the output transfer bins including the leakage in the entire discharge pipe
- PC4.** ensure the output sand is transported from DWUP following the prescribed pump limit
- PC5.** determine if the SMPs output discharge is to be combined with DWUP or divert to the stock yard, considering the information received from DWUP site transfer bin operators
- PC6.** divert the dredge output to the stock yard, in case of problems with the transfer bin, e.g. electrical problem
- PC7.** ensure appropriate discharge from output transfer bin to RPA site of MSP

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** different types of mineral processing processes and the use of relevant equipment for the purpose
- KU2.** the functioning and use of different types of pumps, e.g. slurry pumps, booster pump, water pumps, centrifugal pumps, reciprocating pumps, etc.
- KU3.** the maintenance of pumps, pipelines, pulleys, belts, etc.
- KU4.** different types of feeders, e.g. vibratory feeder, and different types of screens
- KU5.** the principal components of drive mechanisms, lubrication systems, guards, controls, scraper, fire extinguishing equipment, drive belts, hangers/cables, etc.
- KU6.** the safety instructions for conveying systems, fire alarm etc.
- KU7.** the slurry pumping system and associated problems

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- KU8.** the hazards and safety procedures relevant to ore-processing activities, and usage of relevant PPEs
- KU9.** different types of heavy minerals separation processes
- KU10.** the use of various alarms, pumps, screens, valves, conveyor, etc.
- KU11.** the spiral safety procedures
- KU12.** the maintenance and troubleshooting of slurry equipment and hydraulic pumps
- KU13.** different types of classification systems and components including stock tanks, pumping systems. sampling systems, drive systems, lubrication systems, hydraulic systems etc

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** carry out appropriate documentation concerning work
- GS2.** read the equipment manuals, process documents and internal information memos
- GS3.** analyze the available data about the site
- GS4.** enter the relevant information in the online ERP systems
- GS5.** communicate politely and professionally
- GS6.** listen attentively to understand the instructions or information being shared
- GS7.** plan and carry out different activities appropriately, following the task lists and schedules
- GS8.** undertake the work activities as per the applicable task lists and schedules
- GS9.** ensure timely completion of work in compliance with the applicable quality standards
- GS10.** make quick decisions to deal with workplace emergencies
- GS11.** identify potential problems related to work and take appropriate preventive measures
- GS12.** evaluate the possible solutions to work-related problems applying reasoning skills and select the best one
- GS13.** organize the operations and service manuals to allow easy access to them
- GS14.** follow the organizational procedure during emergencies
- GS15.** identify the areas of improvement and take appropriate measures

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Supervise the conveyance of dredged output sand through pipelines</i>	30	50	-	20
PC1. check for the correct functioning of booster pumps added to combined slurry transportation system of mining output	8	12	-	5
PC2. ensure proper monitoring of ampere meter reading and discharge pressure of all booster pumps including pump attached to the output transfer bins	3	6	-	2
PC3. ensure proper monitoring of the water level of all the output transfer bins including the leakage in the entire discharge pipe	3	6	-	2
PC4. ensure the output sand is transported from DWUP following the prescribed pump limit	3	6	-	2
PC5. determine if the SMPs output discharge is to be combined with DWUP or divert to the stock yard, considering the information received from DWUP site transfer bin operators	6	8	-	4
PC6. divert the dredge output to the stock yard, in case of problems with the transfer bin, e.g. electrical problem	4	6	-	3
PC7. ensure appropriate discharge from output transfer bin to RPA site of MSP	3	6	-	2
NOS Total	30	50	-	20

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National Occupational Standards (NOS) Parameters

NOS Code	MIN/N1816
NOS Name	Supervise the conveyance of mining output through slurry pipelines
Sector	Mining
Sub-Sector	Mining Operation
Occupation	Placer Mining
NSQF Level	5.5
Credits	4
Version	1.0
Last Reviewed Date	15/03/2024
Next Review Date	14/03/2027
NSQC Clearance Date	15/03/2024

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MIN/N1810: Follow the applicable health, safety and environmental guidelines (Placer Mining)

Description

This unit is about following the applicable health, safety and environmental guidelines.

Scope

The scope covers the following :

- Maintain personal health and safety
- Assist in hazard management
- Work on equipment and machinery
- Check the first aid box, firefighting and safety equipment
- Report incidents

Elements and Performance Criteria

Maintain personal health and safety

To be competent, the user/individual on the job must be able to:

- PC1.** follow the appropriate hand hygiene measures, i.e., washing hands with EDTA/Soda Ash (Ethylene Diamine Tetra Acetate/Sodium Carbonate), soap and water or alcohol-based hand sanitizers
- PC2.** ensure the plants are provided with a dilution ventilation system to avoid the undue accumulation of chemical vapours in working environments
- PC3.** clean and disinfect the regularly used objects, such as mobile phones, wireless communication systems keys, wallets and work passes
- PC4.** follow the Safety Management Plan (SMP) and Emergency Management Plan (EMP)
- PC5.** follow sanitation measures in the work area, e.g., avoiding eating, drinking, chewing and smoking
- PC6.** use the appropriate PPE, e.g. using an appropriate mask to cover the mouth and nose
- PC7.** identify six directional hazards at the workplace and take decisions accordingly
- PC8.** follow appropriate safety measures to avoid exposure to radiation, e.g. limiting stay in the radiation area and avoiding working with open wounds or exposed body parts in radiation areas
- PC9.** undertake the Take-5 (Personal Risk Assessment) before the commencement of any work
- PC10.** lift objects carefully maintaining the correct body posture to avoid injury
- PC11.** use self-rescue apparatus appropriately when required
- PC12.** work in the assigned work zone in the designated work area
- PC13.** follow the recommended safety measures while working at haul roads, heights, dumps, sump areas, stockyards, near moving parts, etc
- PC14.** wear Thermo luminescent Dosimeter (TLD) Badge for personal monitoring in rare earth extraction plant
- PC15.** use caution signs as applicable and obey safety signs, stickers, and tags

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PC16. identify the environmental impact of operations related to mining, mineral separation plant and rare earth extraction plant and take steps to reduce the impact

PC17. check for contamination of sensitive body parts, e.g. face, eyes etc., and seek immediate medical help

Assist in hazard management

To be competent, the user/individual on the job must be able to:

PC18. label and store chemicals appropriately

PC19. display the MSDS for the hazardous chemicals in storage and handling areas

PC20. educate the personnel at the workplace about the safety requirements and follow the applicable safety guidelines

PC21. check the body and clothing for contamination at the end of radioactive work, and decontaminate and take a shower at the decontamination centre

PC22. maintain regular and open communication with co-workers and the employer

Work on equipment and machinery

To be competent, the user/individual on the job must be able to:

PC23. check and ensure every instrument, apparatus and equipment is AERB approved

PC24. operate the appropriate tools, equipment and machinery with proper training and authorization

Check the first aid box, firefighting, and safety equipment

To be competent, the user/individual on the job must be able to:

PC25. check the availability of first aid kits and fire extinguishers, and ensure they are easily accessible

PC26. ensure the availability of emergency showers and eye washers in the chemical handling, unloading and pumping areas

PC27. monitor the work environment at appropriate intervals for hydrogen chloride and hydrocarbons using portable monitors

PC28. operate the appropriate type of fire extinguishers to control different types of fire at the worksite

Report incidents

To be competent, the user/individual on the job must be able to:

PC29. identify and report potential hazards/ unsafe conditions and practices

PC30. follow the appropriate emergency procedure during emergencies, such as power failures, cyclones, etc.

PC31. report the emergencies and any safety incidents as per the organizational procedure

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. different types of chemical plants and their operations

KU2. the importance of time-keeping, discipline and punctuality

KU3. various chemical processes carried out in a rare earth extraction plant

KU4. the impact of violating applicable safety procedures

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- KU5.** the relevant quality norms
- KU6.** different types of individual rare earth chemical processes and the use of associated equipment
- KU7.** different types of effluent treatment processes and the use of associated equipment
- KU8.** the use of different water purifying and waste treatment agents
- KU9.** the hazards and the appropriate safety measures in acid/ alkali handling activities and usage of relevant PPE
- KU10.** hazards and safety aspects involved in acid/ alkali handling activities and usage of relevant PPE
- KU11.** the internal safety organization and the role of the safety committee, workforce inspector and AERB, etc
- KU12.** the importance of sensitization towards different genders and persons with disabilities (PWD)
- KU13.** the Importance of following infection control policies, '5-S' practices, and waste management as prescribed by regulatory authorities like the Atomic Energy Regulatory Board (AERB), United Nations Development Group (UNDG), etc.
- KU14.** the importance of water/material/energy conservation and management
- KU15.** the importance of Social Management Plan (SMP) and Environment Management Plan (EMP) prepared by the organization, as per DGMS guideline
- KU16.** the importance of wearing own TLD or pocket dosimeter (to be obtained from the Health Physics Unit) in the active area
- KU17.** the importance of staying alert and attentive to the work surroundings
- KU18.** the importance of staying in the assigned work zone in the designated work area
- KU19.** the benefits of making the safety rules part of new employee orientation
- KU20.** the importance of contacting the supervisor or manager for instruction, guidance, or training when required

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** complete appropriate documentation concerning work
- GS2.** make appropriate decisions concerning the area of work with effective reasoning
- GS3.** read and explain manuals, health, and safety instructions, memos, etc
- GS4.** plan and organize work effectively
- GS5.** detect problems at work and use reasoning skills to resolve them promptly and effectively
- GS6.** complete the assigned tasks timely

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Maintain personal health and safety</i>	16	24	-	11
PC1. follow the appropriate hand hygiene measures, i.e., washing hands with EDTA/Soda Ash (Ethylene Diamine Tetra Acetate/Sodium Carbonate), soap and water or alcohol-based hand sanitizers	1	2	-	1
PC2. ensure the plants are provided with a dilution ventilation system to avoid the undue accumulation of chemical vapours in working environments	1	1	-	1
PC3. clean and disinfect the regularly used objects, such as mobile phones, wireless communication systems keys, wallets and work passes	1	1	-	-
PC4. follow the Safety Management Plan (SMP) and Emergency Management Plan (EMP)	1	2	-	-
PC5. follow sanitation measures in the work area, e.g., avoiding eating, drinking, chewing and smoking	1	2	-	-
PC6. use the appropriate PPE, e.g. using an appropriate mask to cover the mouth and nose	1	2	-	-
PC7. identify six directional hazards at the workplace and take decisions accordingly	1	2	-	1
PC8. follow appropriate safety measures to avoid exposure to radiation, e.g. limiting stay in the radiation area and avoiding working with open wounds or exposed body parts in radiation areas	1	2	-	1
PC9. undertake the Take-5 (Personal Risk Assessment) before the commencement of any work	1	2	-	1
PC10. lift objects carefully maintaining the correct body posture to avoid injury	-	1	-	-
PC11. use self-rescue apparatus appropriately when required	1	1	-	1

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC12. work in the assigned work zone in the designated work area	1	1	-	1
PC13. follow the recommended safety measures while working at haul roads, heights, dumps, sump areas, stockyards, near moving parts, etc	1	1	-	1
PC14. wear Thermo luminescent Dosimeter (TLD) Badge for personal monitoring in rare earth extraction plant	1	1	-	1
PC15. use caution signs as applicable and obey safety signs, stickers, and tags	1	1	-	-
PC16. identify the environmental impact of operations related to mining, mineral separation plant and rare earth extraction plant and take steps to reduce the impact	1	1	-	1
PC17. check for contamination of sensitive body parts, e.g. face, eyes etc., and seek immediate medical help	1	1	-	1
<i>Assist in hazard management</i>	5	8	-	4
PC18. label and store chemicals appropriately	1	1	-	1
PC19. display the MSDS for the hazardous chemicals in storage and handling areas	1	2	-	1
PC20. educate the personnel at the workplace about the safety requirements and follow the applicable safety guidelines	1	1	-	-
PC21. check the body and clothing for contamination at the end of radioactive work, and decontaminate and take a shower at the decontamination centre	1	2	-	1
PC22. maintain regular and open communication with co-workers and the employer	1	2	-	1
<i>Work on equipment and machinery</i>	2	4	-	1
PC23. check and ensure every instrument, apparatus and equipment is AERB approved	1	2	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC24. operate the appropriate tools, equipment and machinery with proper training and authorization	1	2	-	1
<i>Check the first aid box, firefighting, and safety equipment</i>	4	10	-	2
PC25. check the availability of first aid kits and fire extinguishers, and ensure they are easily accessible	1	3	-	-
PC26. ensure the availability of emergency showers and eye washers in the chemical handling, unloading and pumping areas	1	2	-	-
PC27. monitor the work environment at appropriate intervals for hydrogen chloride and hydrocarbons using portable monitors	1	2	-	1
PC28. operate the appropriate type of fire extinguishers to control different types of fire at the worksite	1	3	-	1
<i>Report incidents</i>	3	4	-	2
PC29. identify and report potential hazards/ unsafe conditions and practices	1	2	-	-
PC30. follow the appropriate emergency procedure during emergencies, such as power failures, cyclones, etc.	1	1	-	1
PC31. report the emergencies and any safety incidents as per the organizational procedure	1	1	-	1
NOS Total	30	50	-	20

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	MIN/N1810
NOS Name	Follow the applicable health, safety and environmental guidelines (Placer Mining)
Sector	Mining
Sub-Sector	Mining Operation
Occupation	Placer Mining
NSQF Level	4
Credits	1
Version	1.0
Last Reviewed Date	15/03/2024
Next Review Date	14/03/2027
NSQC Clearance Date	15/03/2024

Qualification Pack

MIN/N1811: Work collaboratively and effectively in a team

Description

This unit is about the importance of teamwork in the workplace.

Scope

The scope covers the following :

- Work effectively with co-workers
- Communicate effectively with co-workers
- Practice inclusiveness at work

Elements and Performance Criteria

Work effectively with co-workers

To be competent, the user/individual on the job must be able to:

- PC1.** identify own direct and indirect responsibilities, e.g. duties concerning team members
- PC2.** undertake appropriate activities to promote cooperation and good relationships with co-workers
- PC3.** mentor and assist subordinates in the execution of their work responsibilities
- PC4.** identify possible disruptions to work through coordination with the relevant stakeholders and take appropriate preventive measures
- PC5.** follow the recommended practices to avoid and resolve conflicts at work
- PC6.** interact effectively and cooperate with co-workers
- PC7.** manage workplace conflicts, ensuring no impact on work
- PC8.** identify and take responsibility for own mistakes
- PC9.** follow the organizational policies and procedures
- PC10.** follow the appropriate productivity enhancement measures, e.g. Quality circle, 5s, TPM, Six Sigma, JIT, etc.
- PC11.** exchange feedback with peers and other team members to improve the quality of work

Communicate effectively with co-workers

To be competent, the user/individual on the job must be able to:

- PC12.** follow the organizational policy for the efficient and timely dissemination of information to the authorized personnel
- PC13.** communicate clearly and politely to ensure effective communication with co-workers
- PC14.** practice active listening to understand the conveyed information and instructions
- PC15.** practice openness in work conversations, giving undivided attention to enhance the quality of workplace communication
- PC16.** maintain the professional standard of communication, e.g. avoiding the use of offensive language at work

Practice inclusiveness at work

To be competent, the user/individual on the job must be able to:

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- PC17.** empathize with Persons with Disabilities (PwD)
- PC18.** practice gender equality at work
- PC19.** maintain respectful behaviour and etiquette at work
- PC20.** listen to and acknowledge the feelings, concerns, opinions, and ideas of others

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** the importance of teamwork and the consequence of poor teamwork on project outcomes, timelines, safety, etc
- KU2.** the duties and rights of workers
- KU3.** the importance of creating a healthy and cooperative work environment
- KU4.** the importance of proper and effective communication and the adverse effects of poor communication on quality, timeliness, safety, etc
- KU5.** different modes of communication used at work
- KU6.** the importance and need of supporting co-workers facing problems for the smooth functioning of work
- KU7.** the fundamental concept of gender equality
- KU8.** how to recognize and be sensitive to issues of disability, culture and gender
- KU9.** how to show empathy to PwD appropriately
- KU10.** legislation, policies, and procedures relating to gender sensitivity and cultural diversity including their impact on the area of operation
- KU11.** the appropriate behaviour to be practised with coworkers
- KU12.** the appropriate communication skills and leadership qualities
- KU13.** how to be open to listening to and understanding the other person's point of view

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** complete appropriate documentation concerning work
- GS2.** make appropriate decisions concerning the area of work with effective reasoning
- GS3.** read and explain manuals, health, and safety instructions, memos, etc
- GS4.** plan and organize work effectively
- GS5.** detect problems at work and use reasoning skills to resolve them promptly and effectively
- GS6.** complete the assigned tasks timely

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Work effectively with co-workers</i>	15	30	-	10
PC1. identify own direct and indirect responsibilities, e.g. duties concerning team members	3	5	-	3
PC2. undertake appropriate activities to promote cooperation and good relationships with co-workers	1	2	-	1
PC3. mentor and assist subordinates in the execution of their work responsibilities	1	2	-	1
PC4. identify possible disruptions to work through coordination with the relevant stakeholders and take appropriate preventive measures	3	4	-	2
PC5. follow the recommended practices to avoid and resolve conflicts at work	1	2	-	1
PC6. interact effectively and cooperate with co-workers	1	1	-	-
PC7. manage workplace conflicts, ensuring no impact on work	1	1	-	-
PC8. identify and take responsibility for own mistakes	1	3	-	1
PC9. follow the organizational policies and procedures	1	3	-	-
PC10. follow the appropriate productivity enhancement measures, e.g. Quality circle, 5s, TPM, Six Sigma, JIT, etc.	1	4	-	1
PC11. exchange feedback with peers and other team members to improve the quality of work	1	3	-	-
<i>Communicate effectively with co-workers</i>	9	10	-	5
PC12. follow the organizational policy for the efficient and timely dissemination of information to the authorized personnel	2	3	-	2

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC13. communicate clearly and politely to ensure effective communication with co-workers	1	2	-	-
PC14. practice active listening to understand the conveyed information and instructions	2	1	-	1
PC15. practice openness in work conversations, giving undivided attention to enhance the quality of workplace communication	2	2	-	1
PC16. maintain the professional standard of communication, e.g. avoiding the use of offensive language at work	2	2	-	1
<i>Practice inclusiveness at work</i>	6	10	-	5
PC17. empathize with Persons with Disabilities (PwD)	2	2	-	2
PC18. practice gender equality at work	2	2	-	2
PC19. maintain respectful behaviour and etiquette at work	1	3	-	1
PC20. listen to and acknowledge the feelings, concerns, opinions, and ideas of others	1	3	-	-
NOS Total	30	50	-	20

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	MIN/N1811
NOS Name	Work collaboratively and effectively in a team
Sector	Mining
Sub-Sector	Mining Operation
Occupation	Placer Mining
NSQF Level	4
Credits	1
Version	1.0
Last Reviewed Date	15/03/2024
Next Review Date	14/03/2027
NSQC Clearance Date	15/03/2024

Qualification Pack

DGT/VSQ/N0103: Employability Skills (90 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Career Development & Goal Setting
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

- PC1.** understand the significance of employability skills in meeting the current job market requirement and future of work
- PC2.** identify and explore learning and employability relevant portals
- PC3.** research about the different industries, job market trends, latest skills required and the available opportunities

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

- PC4.** recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.
- PC5.** follow environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

- PC6.** recognize the significance of 21st Century Skills for employment

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- PC7.** practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life
- PC8.** adopt a continuous learning mindset for personal and professional development

Basic English Skills

To be competent, the user/individual on the job must be able to:

- PC9.** use basic English for everyday conversation in different contexts, in person and over the telephone
- PC10.** read and understand routine information, notes, instructions, mails, letters etc. written in English
- PC11.** write short messages, notes, letters, e-mails etc. in English

Career Development & Goal Setting

To be competent, the user/individual on the job must be able to:

- PC12.** identify career goals based on the skills, interests, knowledge, and personal attributes
- PC13.** prepare a career development plan with short- and long-term goals

Communication Skills

To be competent, the user/individual on the job must be able to:

- PC14.** follow verbal and non-verbal communication etiquette while communicating in professional and public settings
- PC15.** use active listening techniques for effective communication
- PC16.** communicate in writing using appropriate style and format based on formal or informal requirements
- PC17.** work collaboratively with others in a team

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

- PC18.** communicate and behave appropriately with all genders and PwD
- PC19.** escalate any issues related to sexual harassment at workplace according to POSH Act

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

- PC20.** identify and select reliable institutions for various financial products and services such as bank account, debit and credit cards, loans, insurance etc.
- PC21.** carry out offline and online financial transactions, safely and securely, using various methods and check the entries in the passbook
- PC22.** identify common components of salary and compute income, expenses, taxes, investments etc
- PC23.** identify relevant rights and laws and use legal aids to fight against legal exploitation

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

- PC24.** operate digital devices and use their features and applications securely and safely
- PC25.** carry out basic internet operations by connecting to the internet safely and securely, using the mobile data or other available networks through Bluetooth, Wi-Fi, etc.
- PC26.** display responsible online behaviour while using various social media platforms

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- PC27.** create a personal email account, send and process received messages as per requirement
- PC28.** carry out basic procedures in documents, spreadsheets and presentations using respective and appropriate applications
- PC29.** utilize virtual collaboration tools to work effectively

Entrepreneurship

To be competent, the user/individual on the job must be able to:

- PC30.** identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research
- PC31.** develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion
- PC32.** identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity

Customer Service

To be competent, the user/individual on the job must be able to:

- PC33.** identify different types of customers and ways to communicate with them
- PC34.** identify and respond to customer requests and needs in a professional manner
- PC35.** use appropriate tools to collect customer feedback
- PC36.** follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

- PC37.** create a professional Curriculum vitae (Résumé)
- PC38.** search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively
- PC39.** apply to identified job openings using offline /online methods as per requirement
- PC40.** answer questions politely, with clarity and confidence, during recruitment and selection
- PC41.** identify apprenticeship opportunities and register for it as per guidelines and requirements

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** need for employability skills and different learning and employability related portals
- KU2.** various constitutional and personal values
- KU3.** different environmentally sustainable practices and their importance
- KU4.** Twenty first (21st) century skills and their importance
- KU5.** how to use English language for effective verbal (face to face and telephonic) and written communication in formal and informal set up
- KU6.** importance of career development and setting long- and short-term goals
- KU7.** about effective communication
- KU8.** POSH Act
- KU9.** Gender sensitivity and inclusivity
- KU10.** different types of financial institutes, products, and services

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- KU11.** components of salary and how to compute income and expenditure
- KU12.** importance of maintaining safety and security in offline and online financial transactions
- KU13.** different legal rights and laws
- KU14.** different types of digital devices and the procedure to operate them safely and securely
- KU15.** how to create and operate an e- mail account
- KU16.** use applications such as word processors, spreadsheets etc.
- KU17.** how to identify business opportunities
- KU18.** types and needs of customers
- KU19.** how to apply for a job and prepare for an interview
- KU20.** apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read and write different types of documents/instructions/correspondence in English and other languages
- GS2.** communicate effectively using appropriate language in formal and informal settings
- GS3.** behave politely and appropriately with all to maintain effective work relationship
- GS4.** how to work in a virtual mode, using various technological platforms
- GS5.** perform calculations efficiently
- GS6.** solve problems effectively
- GS7.** pay attention to details
- GS8.** manage time efficiently
- GS9.** maintain hygiene and sanitization to avoid infection

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. understand the significance of employability skills in meeting the current job market requirement and future of work	-	-	-	-
PC2. identify and explore learning and employability relevant portals	-	-	-	-
PC3. research about the different industries, job market trends, latest skills required and the available opportunities	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC4. recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.	-	-	-	-
PC5. follow environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	1	3	-	-
PC6. recognize the significance of 21st Century Skills for employment	-	-	-	-
PC7. practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life	-	-	-	-
PC8. adopt a continuous learning mindset for personal and professional development	-	-	-	-
<i>Basic English Skills</i>	3	4	-	-
PC9. use basic English for everyday conversation in different contexts, in person and over the telephone	-	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. read and understand routine information, notes, instructions, mails, letters etc. written in English	-	-	-	-
PC11. write short messages, notes, letters, e-mails etc. in English	-	-	-	-
<i>Career Development & Goal Setting</i>	1	2	-	-
PC12. identify career goals based on the skills, interests, knowledge, and personal attributes	-	-	-	-
PC13. prepare a career development plan with short- and long-term goals	-	-	-	-
<i>Communication Skills</i>	2	2	-	-
PC14. follow verbal and non-verbal communication etiquette while communicating in professional and public settings	-	-	-	-
PC15. use active listening techniques for effective communication	-	-	-	-
PC16. communicate in writing using appropriate style and format based on formal or informal requirements	-	-	-	-
PC17. work collaboratively with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	1	-	-
PC18. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC19. escalate any issues related to sexual harassment at workplace according to POSH Act	-	-	-	-
<i>Financial and Legal Literacy</i>	2	3	-	-
PC20. identify and select reliable institutions for various financial products and services such as bank account, debit and credit cards, loans, insurance etc.	-	-	-	-
PC21. carry out offline and online financial transactions, safely and securely, using various methods and check the entries in the passbook	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC22. identify common components of salary and compute income, expenses, taxes, investments etc	-	-	-	-
PC23. identify relevant rights and laws and use legal aids to fight against legal exploitation	-	-	-	-
<i>Essential Digital Skills</i>	3	5	-	-
PC24. operate digital devices and use their features and applications securely and safely	-	-	-	-
PC25. carry out basic internet operations by connecting to the internet safely and securely, using the mobile data or other available networks through Bluetooth, Wi-Fi, etc.	-	-	-	-
PC26. display responsible online behaviour while using various social media platforms	-	-	-	-
PC27. create a personal email account, send and process received messages as per requirement	-	-	-	-
PC28. carry out basic procedures in documents, spreadsheets and presentations using respective and appropriate applications	-	-	-	-
PC29. utilize virtual collaboration tools to work effectively	-	-	-	-
<i>Entrepreneurship</i>	2	3	-	-
PC30. identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research	-	-	-	-
PC31. develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion	-	-	-	-
PC32. identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity	-	-	-	-
<i>Customer Service</i>	1	2	-	-
PC33. identify different types of customers and ways to communicate with them	-	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC34. identify and respond to customer requests and needs in a professional manner	-	-	-	-
PC35. use appropriate tools to collect customer feedback	-	-	-	-
PC36. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	2	3	-	-
PC37. create a professional Curriculum vitae (Résumé)	-	-	-	-
PC38. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively	-	-	-	-
PC39. apply to identified job openings using offline /online methods as per requirement	-	-	-	-
PC40. answer questions politely, with clarity and confidence, during recruitment and selection	-	-	-	-
PC41. identify apprenticeship opportunities and register for it as per guidelines and requirements	-	-	-	-
NOS Total	20	30	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0103
NOS Name	Employability Skills (90 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	5
Credits	3
Version	1.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. Criteria for assessment for each Qualification Pack will be created by the Sector Skill Council. Each Element/ Performance Criteria (PC) will be assigned marks proportional to its importance in NOS. SSC will also lay down proportion of marks for Theory and Skills Practical for each Element/ PC.
2. The assessment for the theory part will be based on knowledge bank of questions created by the SSC.
3. Assessment will be conducted for all compulsory NOS, and where applicable, on the selected elective/option NOS/set of NOS.
4. Individual assessment agencies will create unique question papers for theory part for each candidate at each examination/training center (as per assessment criteria below).
5. Individual assessment agencies will create unique evaluations for skill practical for every student at each examination/ training center based on these criteria.
6. To pass the Qualification Pack assessment, every trainee should score the Recommended Pass % aggregate for the QP.
7. In case of unsuccessful completion, the trainee may seek reassessment on the Qualification Pack.

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Minimum Aggregate Passing % at QP Level : 70

(**Please note:** Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
MIN/N1814.Supervise excavation, accumulation, loading and transportation of raw sand	30	50	0	20	100	30
MIN/N1815.Supervise the safety of dredging operations at DWUP	30	50	0	20	100	30
MIN/N1816.Supervise the conveyance of mining output through slurry pipelines	30	50	0	20	100	25
MIN/N1810.Follow the applicable health, safety and environmental guidelines (Placer Mining)	30	50	0	20	100	5
MIN/N1811.Work collaboratively and effectively in a team	30	50	0	20	100	5
DGT/VSQ/N0103.Employability Skills (90 Hours)	20	30	-	-	50	5
Total	170	280	-	100	550	100

Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training

Qualification Pack

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

Qualification Pack

Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.